

ISC COMPUTER SCIENCE – TEST II (60 Marks)

Syllabus: Basic I/O, Packages, Logic Gates, Recursion, Arrays & Strings

SECTION A – MCQs (20 × 1 = 20 marks)

1. Which keyword is used to define a package in Java?
a) import **b) package** c) namespace d) module
2. The output of an AND gate is 1 when:
a) All inputs are 1 b) Any input is 1 c) Inputs differ d) All inputs are 0
3. A recursive function must always contain:
a) Loop b) Return statement **c) Base case** d) Parameter
4. Arrays in Java are:
a) Dynamically resized **b) Fixed-sized** c) Only for integers d) Two-dimensional only
5. Which of the following is NOT a basic logic gate?
a) AND b) NOT **c) XOR** d) OR
6. String objects in Java are:
a) Mutable b) Immutable c) Temporary d) Dynamic memory blocks
7. The NOT gate outputs 1 when input is:
a) 0 b) 1 c) Either 0 or 1 d) None
8. `import java.util.*;` means:
a) Import all Java classes
b) Import only Scanner
c) Import all classes of util package
d) Import the package util itself
9. Recursion internally uses which data structure?
a) Queue **b) Stack** c) Tree d) Linked list
10. What will XOR (1, 1) output?
a) 0 b) 1 c) -1 d) Same as OR
11. Which array index is invalid?
a) `arr[0]` b) `arr[1]` **c) `arr[n]` for size n** d) `arr[n-1]`
12. Packages help in:
a) Slowing execution
b) Grouping related classes
c) Storing loops
d) Removing errors
13. Recursion is most suitable for:
a) Repetitive hardware tasks
b) Problems reducible to smaller subproblems
c) Simple printing tasks
d) File writing

14. Which symbol represents OR gate?
a) + b) . c) $\oplus$ d) ' **a) +**
15. A two-dimensional array is also called a:
a) Matrix b) Tree c) Queue d) Line
16. length in arrays refers to:
a) Memory size **b) Number of elements**
c) Last index number d) None
17. The truth table of NOT gate has how many inputs?
a) 0 **b) 1** c) 2 d) Variable
18. Which method returns the length of a String?
a) size() b) len() **c) length()** d) get()
19. A package can contain:
a) Only methods b) Only classes
c) Classes and interfaces d) Only variables
20. Breaking a problem into smaller subproblems refers to:
a) Compilation **b) Recursion** c) Repetition d) Overloading
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SECTION B – Assertion–Reasoning (5 × 2 = 10 marks)

(Choose: a) Both A & R true, R explains A
b) Both true, R doesn't explain A
c) A true, R false
d) A false, R true)

21. **A:** NAND gate is called a universal gate.
R: Any digital circuit can be built using only NAND gates.
22. **A:** A recursive function without a base case leads to stack overflow.
R: Recursion uses the function call stack repeatedly.
23. **A:** String objects in Java cannot be modified.
R: The String class in Java is immutable.
24. **A:** Arrays in Java are passed by reference.
R: Array variables store memory addresses of actual array data.
25. **A:** The OR gate outputs 1 when at least one input is 1.
R: OR gate performs logical addition.
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SECTION C – One-sentence Answers (5 × 2 = 10 marks)

26. Define recursion.
27. What is a package?

- 28. What is meant by the term "truth table"?
 - 29. What is an array index?
 - 30. What does the XOR gate do when both inputs are the same?
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SECTION D – Programming Questions (4 × 5 = 20 marks)

(No file-handling programs included)

- 31. Write a recursive function in Java to find the factorial of a number.
- 32. Write a Java program to input 10 elements in an array and display all prime numbers from the array.
- 33. Write a Java program to count the number of vowels, consonants, digits and spaces in a given string.
- 34. Create a package named `mathops` containing a class `Series` with a method to generate the first `n` Fibonacci numbers. Write another class to access and display it.
- 35. Write a Java program to find the largest and smallest word in a given sentence.